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Machine-learning-based detection of fake videos

Abstract

A method for training a model for classifying videos as real or fake can include generating image tiles and audio data segments from an input video, generating a sequence of image embeddings based on the image tiles using a visual encoder and a sequence of audio embeddings based on the audio data segments using an audio encoder, transforming, using a V2A network, a first subset of the sequence of image embeddings into synthetic audio embeddings, transforming, using an A2V network, a first subset of the sequence of audio embeddings into synthetic image embeddings, updating the sequence of image embeddings by using the synthetic image embeddings, updating the sequence of audio embeddings using the synthetic audio embeddings, training the encoders and the networks using the updated sequences of image embeddings and audio embeddings, and training a classifier using the trained encoders and the trained networks.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/600,581, filed Nov. 17, 2023, the entire contents of which are hereby incorporated by reference herein.

FIELD

(1) The present disclosure relates to techniques for detecting fake (e.g., AI-generated) videos.

BACKGROUND

(2) Generative AI technology has enabled the creation of rich, high-quality multimedia content. However, the technology is increasingly being leveraged to defraud, defame, and spread disinformation. The malicious use of generative AI technology therefore poses a major societal threat.

(3) AI-generated videos can be particularly misleading. These videos can include AI-generated audio and real visuals, real audio and AI-generated visuals, or both AI-generated audio and AI-generated visuals. Correspondences between a video's audio data and visual data can indicate whether the video is fake or real. However, many existing techniques for detecting AI-generated videos focus on data of a particular modality (e.g., only audio data or only visual data) and, as a result, are frequently unable to identify as fake videos with real data of that modality but fake data of the other modality. Other techniques use supervised learning to train analytic models to classify videos as real or fake by implicitly capturing audio-visual correspondences. The focus of such models is usually restricted to the specific correspondences present in the training data set, which may cause the models to overlook correspondences that can help detect unseen AI-generated videos.

SUMMARY

(4) Provided are machine-learning-based techniques for training a model to detect fake (e.g., AI-generated or deepfake) videos. The model can include a visual encoder, an audio encoder, an audio-to-visual (A2V) network, a visual-to-audio (V2A) network, and a classifier for classifying videos as real or fake. The encoders and the networks may be trained on real videos by first using the encoders to generate sequences of image embeddings and audio embeddings from image and audio data extracted from input videos and then replacing a subset of each embedding sequence with synthetic embeddings generated by one of the networks using a corresponding subset of the embeddings for the opposite modality. Specifically, a subset of the sequence of image embeddings may be replaced by synthetic image embeddings generated by the A2V network based on a corresponding subset of the audio embeddings and a subset of the sequence of audio embeddings may be replaced by synthetic audio embeddings generated by the V2A network based on a corresponding subset of the image embeddings. From these “cross-modal” representations produced using embeddings from both the audio and visual modalities, the encoders and the networks can learn to capture intrinsic correspondences between the audio and visual modalities in real videos.

(5) Once the encoders and the networks are trained, they may be used to produce cross-modal representations of videos to be classified as real or fake. Training the encoders and the networks using real videos may ensure that, for real videos, the cross-modal representations generated by the encoders and the networks display high audio-visual cohesion and, for fake videos (e.g., videos with fake images, videos with fake audio, or videos with fake images and fake audio), the cross-modal representations generated by the encoders and the network display low audio-visual cohesion. The differences in audio-visual cohesion in representations of real videos generated by

the encoders and the networks and in representations of fake videos generated by the encoders and the networks can be exploited to train the classifier to distinguish between real videos and fake videos with high accuracy.

(6) The disclosed techniques provide numerous technical advantages. In various embodiments, the techniques may improve the functioning of a computer by reducing processing power, battery usage, and memory requirements associated with detecting fake videos. The provided cross-modal learning method may produce trained models with broad focuses that can accurately analyze a wide variety of videos. In particular, classifiers trained using the provided cross-modal learning method may be capable of interpreting a range of audio-visual correspondences and, as a result, may perform accurately on videos having fake audio, videos having fake visuals, and videos having both fake audio and fake visuals.

(7) A method for training a model comprising a visual encoder, an audio encoder, an audio-to-visual (A2V) network, a visual-to-audio (V2A) network, and a classifier for classifying videos as real or fake can comprise generating a sequence of image tiles from image data from an input video and generating a plurality of data segments representing audio data from the input video. A sequence of image embeddings can be generated based on the sequence of image tiles using the visual encoder. Similarly, a sequence of audio embeddings can be generated based on the sequence of data segments using the audio encoder. The V2A network can be used to transform a first subset of the sequence of image embeddings into one or more synthetic audio embeddings and the A2V network can be used to transform a first subset of the sequence of audio embeddings into one or more synthetic image embeddings. The sequence of image embeddings can then be updated by replacing a second subset of the sequence of image embeddings with the one or more synthetic image embeddings. Likewise, the sequence of audio embeddings can be updated by replacing a second subset of the sequence of audio embeddings with the one or more synthetic audio embeddings. The visual encoder, the audio encoder, the V2A network, and the A2V network can be trained based on the updated sequence of image embeddings and the updated sequence of audio embeddings. The classifier can be trained to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network, wherein the classifier is configured to receive image embeddings for the videos from the trained visual encoder, audio embeddings for the videos from the trained audio encoder, synthetic image embeddings for the videos from the trained A2V network, and synthetic audio embeddings for the videos from the trained V2A network.

(8) The first subset of the sequence of image embeddings can include half of the image embeddings, and the first subset of the sequence of audio embeddings can include half of the audio embeddings. In some embodiments, the first subset of the sequence of image embeddings and the first subset of the sequence of audio embeddings are randomly selected. The first subset of the sequence of image embeddings can correspond to a first set of time points in the input video, and the first subset of the sequence of audio embeddings can correspond to a second set of time points in the input video different from the first set of time points. The second subset of the sequence of image embeddings can correspond to the second set of time points, and the second subset of the sequence of audio embeddings can correspond to the first set of time points.

(9) Training the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated sequence of audio embeddings can include decoding the updated sequence of image embeddings to produce a reconstruction of the sequence of image tiles and decoding the updated sequence of audio embeddings to produce a reconstruction of the plurality of data segments. The updated sequence of image embeddings may be decoded using a visual decoder and the updated sequence of audio embeddings may be decoded using an audio decoder. In some embodiments, training the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated sequence of audio embeddings further comprises computing a dual-objective loss, wherein

a first objective of the dual-objective loss depends on the sequence of audio embeddings and the sequence of image embeddings and a second objective of the dual-objective loss depends on the sequence of image tiles, the plurality of data segments, the reconstruction of the sequence of image tiles, and the reconstruction of the plurality of data segments.

(10) Training the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network can include generating a second sequence of image tiles from image data from a labeled training video comprising a label indicating whether the labeled training video is real or fake, generating a second plurality of data segments representing audio data from the labeled training video, generating a second sequence of image embeddings based on the second sequence of image tiles using the trained visual encoder, generating a second sequence of audio embeddings based on the second sequence of data segments using the trained audio encoder, transforming, using the trained V2A network, the second sequence of image embeddings into a sequence of synthetic audio embeddings, transforming, using the trained A2V network, the second sequence of audio embeddings into a sequence synthetic image embeddings, concatenating the second sequence of image embeddings and the sequence of synthetic image embeddings to produce a combined sequence of image embeddings, concatenating the second sequence of audio embeddings and the sequence of synthetic audio embeddings to produce a combined sequence of audio embeddings, and classifying the labeled training video as real or fake based on the combined sequence of audio embeddings and the combined sequence of image embeddings. Training the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network can further comprise computing a cross-entropy loss objective using label indicating whether the labeled training video is real or fake. The classifier can include an audio mode patch reduction network, a visual mode patch reduction network, and a classifier head.

(11) A number of image tiles in the sequence of image tiles and a number of data segments in the sequence of data segments can be determined based on a sampling frequency of the image data, a sampling frequency of the audio data, and a time duration of the input video. The input video may show a human face.

(12) In some embodiments, the method further comprises providing the trained model with a second input video and classifying the second input video as real or fake using the trained model. The second input video can include real audio data and AI-generated image data, real image data and AI-generated audio data, or AI-generated image data and AI-generated audio data.

(13) A system for training a model comprising a visual encoder, an audio encoder, an audio-to-visual (A2V) network, a visual-to-audio (V2A) network, and a classifier for classifying videos as real or fake can include one or more processors configured to: generate a sequence of image tiles from image data from an input video, generate a plurality of data segments representing audio data from the input video, generate a sequence of image embeddings based on the sequence of image tiles using the visual encoder, generate a sequence of audio embeddings based on the sequence of data segments using the audio encoder, transform, using the V2A network, a first subset of the sequence of image embeddings into one or more synthetic audio embeddings, transform, using the A2V network, a first subset of the sequence of audio embeddings into one or more synthetic image embeddings, update the sequence of image embeddings by replacing a second subset of the sequence of image embeddings with the one or more synthetic image embeddings, update the sequence of audio embeddings by replacing a second subset of the sequence of audio embeddings with the one or more synthetic audio embeddings, train the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated sequence of audio embeddings, train the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network, wherein the classifier is configured to receive image embeddings for the videos from the

trained visual encoder, audio embeddings for the videos from the trained audio encoder, synthetic image embeddings for the videos from the trained A2V network, and synthetic audio embeddings for the videos from the trained V2A network.

(14) A non-transitory computer readable storage medium storing instructions for training a model comprising a visual encoder, an audio encoder, an audio-to-visual (A2V) network, a visual-to-audio (V2A) network, and a classifier for classifying videos as real or fake that, when executed by one or more processors of a computer system, can cause the computer system to: generate a sequence of image tiles from image data from an input video, generate a plurality of data segments representing audio data from the input video, generate a sequence of image embeddings based on the sequence of image tiles using the visual encoder, generate a sequence of audio embeddings based on the sequence of data segments using the audio encoder, transform, using the V2A network, a first subset of the sequence of image embeddings into one or more synthetic audio embeddings, transform, using the A2V network, a first subset of the sequence of audio embeddings into one or more synthetic image embeddings, update the sequence of image embeddings by replacing a second subset of the sequence of image embeddings with the one or more synthetic image embeddings, update the sequence of audio embeddings by replacing a second subset of the sequence of audio embeddings with the one or more synthetic audio embeddings, train the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated sequence of audio embeddings, train the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network, wherein the classifier is configured to receive image embeddings for the videos from the trained visual encoder, audio embeddings for the videos from the trained audio encoder, synthetic image embeddings for the videos from the trained A2V network, and synthetic audio embeddings for the videos from the trained V2A network.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.
- (2) The following figures show various systems, methods, and apparatuses detecting fake videos. The systems, methods, and apparatuses shown in the figures may have any one or more of the characteristics described herein.
- (3) FIG. 1 shows a block diagram of a system for detecting fake videos, according to some embodiments.
- (4) FIG. 2 shows a method for training an audio-to-visual (A2V) network and a visual-to-audio (V2A) network to acquire cross-modal representations of videos, according to some embodiments.
- (5) FIG. 3A shows a schematic of a process for generating sequences of audio embeddings and visual embeddings from an input video, according to some embodiments.
- (6) FIG. 3B shows a schematic of a complementary masking process, according to some embodiments.
- (7) FIG. 3C shows a schematic of a feature fusion process for generating cross-modal representations, according to some embodiments.
- (8) FIG. 3D shows a schematic of a process for reconstructing input video data from cross-modal representations, according to some embodiments.
- (9) FIG. 4 shows a method for training a classifier to detect fake videos, according to some embodiments.
- (10) FIG. 5A shows a schematic of first portion of a process for training a classifier to detect fake

videos, according to some embodiments.

(11) FIG. 5B shows a schematic of second portion of a process for training a classifier to detect fake videos, according to some embodiments.

(12) FIG. 5C shows a schematic of third portion of a process for training a classifier to detect fake videos, according to some embodiments.

(13) FIG. 5D shows a schematic of fourth portion of a process for training a classifier to detect fake videos, according to some embodiments.

(14) FIG. 6 shows a computer system, according to some embodiments.

(15) FIG. 7 shows t-Stochastic Neighbor Embedding plots for an example dataset comprising real and fake videos.

(16) FIG. 8A shows example data illustrating the AUC scores as a function of different levels of intensities for various visual perturbations evaluated on a test data set.

(17) FIG. 8B shows example data illustrating the AUC scores as a function of different levels of intensities for various visual perturbations evaluated on a test data set.

(18) FIG. 9 shows example data illustrating the variation of AUC and AP scores as a function of different levels of intensities for various audio perturbations evaluated on a test data set.

DETAILED DESCRIPTION

(19) Provided are machine-learning-based techniques for training a model to detect fake (e.g., AI-generated or deepfake) videos. The model can include a visual encoder, an audio encoder, an audio-to-visual (A2V) network, a visual-to-audio (V2A) network, and a classifier for classifying videos as real or fake. The encoders and the networks may be trained on real videos by first using the encoders to generate sequences of image embeddings and audio embeddings from image and audio data extracted from input videos and then replacing a subset of each embedding sequence with synthetic embeddings generated by one of the networks using a corresponding subset of the embeddings for the opposite modality. Specifically, a subset of the sequence of image embeddings may be replaced by synthetic image embeddings generated by the A2V network based on a corresponding subset of the audio embeddings and a subset of the sequence of audio embeddings may be replaced by synthetic audio embeddings generated by the V2A network based on a corresponding subset of the image embeddings. From these “cross-modal” representations produced using embeddings from both the audio and visual modalities, the encoders and the networks can learn to capture intrinsic correspondences between the audio and visual modalities in real videos.

(20) Once the encoders and the networks are trained, they may be used to produce cross-modal representations of videos to be classified as real or fake. Training the encoders and the networks using real videos may ensure that, for real videos, the cross-modal representations generated by the encoders and the networks display high audio-visual cohesion and, for fake videos (e.g., videos with fake images, videos with fake audio, or videos with fake images and fake audio), the cross-modal representations generated by the encoders and the network display low audio-visual cohesion. The differences in audio-visual cohesion in representations of real videos generated by the encoders and the networks and in representations of fake videos generated by the encoders and the networks can be exploited to train the classifier to distinguish between real videos and fake videos with high accuracy.

(21) The following description sets forth exemplary methods, parameters, and the like. It should be recognized, however, that such description is not intended as a limitation on the scope of the present disclosure but is instead provided as a description of exemplary embodiments.

(22) Although the following description uses terms “first,” “second,” etc. to describe various elements, these elements should not be limited by the terms. These terms are only used to distinguish one element from another. For example, a first graphical representation could be termed a second graphical representation, and, similarly, a second graphical representation could be termed a first graphical representation, without departing from the scope of the various described

embodiments. The first graphical representation and the second graphical representation are both graphical representations, but they are not the same graphical representation.

(23) The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(24) The term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

(25) The provided methods, systems, apparatuses, and non-transitory computer readable storage media may identify videos as real or fake using a classifier that has been trained using cross-modal video representations generated by models that have learned audio-visual correspondences inherent to real videos. These models may be trained via a self-supervised learning paradigm that employs a contrastive learning objective and a complementary masking and fusion strategy that sits within an autoencoding objective. The complementary masking and fusion strategy may take uni-modal audio and visual embeddings and systematically mask them to force the learning of advanced embeddings via reconstruction. To instill cross-modal dependency, tokens from one modality may be used to learn the masked embeddings of the other modality via cross-modal token conversion networks. Training the encoders and cross-modal networks on real videos may compel the models to learn dependencies between real audio and corresponding real visual data. The high audio-visual correspondences in the representations of real videos generated by the trained models may be leveraged to train the classifier to distinguish between real and fake videos by exploiting the lack of audio-visual cohesion in synthesized video samples.

(26) An exemplary system for detecting fake videos is illustrated in FIG. 1. The system may include trained audio and visual encoders **102**, trained audio-to-visual (A2V) and visual-to-audio (V2A) networks **104**, and a trained classifier **106**. Audio and visual encoders **102** and A2V and V2A networks **104** may generate cross-modal representations of videos using audio-visual correspondences learned from training on real videos. Classifier **106** may have been trained using cross-modal representations produced by audio and visual encoders **102** and A2V and V2A networks **104** to classify a video as real or fake based on the video's audio-visual cohesion.

(27) To detect whether a given video **108** is real or fake, audio data and visual data from video **108** may be provided to trained audio and visual encoders **102**, which may generate audio embeddings from the audio data and image embeddings from the visual data. The audio and image embeddings may then be passed to trained A2V and V2A networks **104**, at which point the trained A2V network may synthesize a set of image embeddings using the audio embeddings generated by the trained audio encoder and the trained V2A network may synthesize a set of audio embeddings using the image embeddings generated by the trained visual encoder. The synthetic image embeddings generated by the A2V network may be concatenated with the image embeddings generated by the visual encoder while preserving the temporal position of each embedding in its respective sequence. Similarly, the synthetic audio embeddings generated by the V2A network may be concatenated with the audio embeddings generated by the audio encoder while preserving the temporal position of each embedding in its respective sequence. These concatenated sets of

embeddings may then be provided as input to classifier **106**, which may output either an indication **112a** that the video is real or an indication **112b** that the video is fake.

(28) FIG. 2 provides an exemplary method **200** for training an audio encoder, a visual encoder, an audio-to-visual (A2V) network, and a visual-to-audio (V2A) network. Method **200** may be executed using a computer system and may produce trained audio and visual encoders and trained A2V and V2A networks such as audio and visual encoders **102** and A2V and V2A networks **104** of system **100** shown in FIG. 1. The computer system used to perform method **200** may, for example, comprise one or more electronic devices implementing a software platform. In other examples, the computer system may be a client-server system, in which case the blocks of method **200** may be divided up between the server and a client device, between the server and multiple client devices, using only a client device, or using only multiple client devices.

(29) In various embodiments of method **200**, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the blocks of method **200**. Accordingly, the operations as illustrated (and described in greater detail below) are exemplary by nature and, as such, should not be viewed as limiting.

(30) A preprocessing stage may be performed prior to executing method **200**. In the preprocessing stage, an input video may be processed to extract image data and audio data. The image data and the audio data may respectively comprise visual frames and audio waveforms extracted from the input video at predetermined sampling rates. In various embodiments, the sampling rate at which visual frames are extracted from the input video is between 24 and 120 fps, for example 24 fps, 30 fps, 60 fps, or 120 fps. The sampling rate at which audio waveforms are extracted from the input video may be between 8 and 48 kHz, for example 8 kHz, 16 kHz, 22.1 kHz, 44.1 kHz, or 48 kHz.

(31) The extracted audio waveforms that make up the audio data may be converted into spectrograms (e.g., log-mel spectrograms with L frequency bins). Optionally, the extracted image data and the extracted audio data can be further processed to remove data that has minimal impact on audio-visual correspondence. For example, background regions of a visual frame in the image data that contribute minimally to the video's audio may be cropped or otherwise eliminated. If, for instance, the input video shows a human face while the human is speaking, the visual frames that make up the image data for said video may be cropped to select the facial regions and eliminate the background. This may be accomplished using any suitable technique, for example a facial recognition toolbox such as the PyTorch toolbox FaceX-Zoo.

(32) After the image data and the audio data have been extracted and processed, a sequence of image tiles may be generated from the image data (step **202** of method **200**) and a plurality of audio data segments may be generated from the audio data (step **204** of method **200**). That is, for an input video x with a total time duration T that has audio data components $x_{\text{sub.a}} \in \text{custom character}$ $_{\text{sup.T.sup.a.sup.} \times L}$ and image data components $x_{\text{sub.v}} \in \text{custom character}$ $_{\text{sup.T.sup.v.sup.} \times C \times H \times W}$, a set of N equal temporal audio data segments $X_{\text{sub.a}} =$

$\{X_{\text{sub.a},t_{\text{sub.i}}}\}_{\text{sub.i}=1}^{\text{sup.N}}$ may be generated from the audio data and a set of N equal temporal image tiles $X_{\text{sub.v}} = \{X_{\text{sub.v},t_{\text{sub.i}}}\}_{\text{sub.i}=1}^{\text{sup.N}}$ may be generated from the image data.

$(T_{\text{sub.a}}, L)$ may denote the number of audio frames and the number of frequency bins in the spectrograms for the audio waveforms, respectively, while $(T_{\text{sub.v}}, H, W, C)$ may denote the number of visual frames, height, width, and number of channels in the image data, respectively. $T_{\text{sub.a}}$ and $T_{\text{sub.v}}$ may be such that:

(33) $T_v \cdot n_v = T_a \cdot n_a = T$ (Equation1)

where $n_{\text{sub.v}}$ is the sampling rate at which image frames were extracted from the input video and $n_{\text{sub.a}}$ is the sampling rate at which audio waveforms were extracted from the input video. To generate $X_{\text{sub.a}}$, the audio data components $x_{\text{sub.a}}$ may be tokenized using $P \times P$ non-overlapping 2D patches (e.g., similar to Audio-MAE, described in: Po-Yao Huang, Hu Xu, Juncheng Li, Alexei

Baevski, Michael Auli, Wojciech Galuba, Florian Metze, and Christoph Feichtenhofer. Masked autoencoders that listen. In *Advances in Neural Information Processing Systems*, 35:28708-28720, (2022). For example, the audio data components may be tokenized using 16×16 , 14×14 , or 32×32 non-overlapping 2D patches. To generate $X_{\text{sub.v}}$, the image data components $x_{\text{sub.v}}$ may be tokenized using $2 \times P \times P$ (e.g., $2 \times 16 \times 16$) non-overlapping 3D spatio-temporal patches (e.g., similar to MARLIN, described in: Zhixi Cai, Shreya Ghosh, Kalin Stefanov, Abhinav Dhall, Jianfei Cai, Hamid Rezatofighi, Reza Haffari, and Munawar Hayat. Marlin: Masked autoencoder for facial video representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1493-621 1504, 2023).

(34) The number N of temporal slices may be determined empirically. In some embodiments, the number N of temporal slices is greater than or equal to 2, 4, 6, 8, 10, 12, 14, 16, 18, or 20. In other embodiments, the number N of temporal slices is less than or equal to 1000, 500, 100, 90, 80, 70, 60, or 50. Tokenization of the audio data components and the image data components may be performed such that the temporal correspondence of the audio data segments and the image tiles is preserved, i.e., such that $x_{\text{sub.a.t.sub.i}}$ and $x_{\text{sub.v.t.sub.i}}$ correspond to the same time interval.

(35) After the sequence of image tiles is generated, a sequence of image embeddings may be generated (step **206** of method **200**). Likewise, after the plurality of audio data segments is generated, a sequence of audio embeddings may be generated (step **208** of method **200**). The sequence of image embeddings may be generated by a visual encoder $E_{\text{sub.v}}$ and the sequence of audio embeddings may be generated by an audio encoder $E_{\text{sub.a}}$. The visual encoder $E_{\text{sub.v}}$ may encode the image tiles $X_{\text{sub.v}}$ and output uni-modal features v , where:

$$(36) \quad v = \{v_{t_i}\}_{i=1}^N = E_v(X_v + \text{pos}_v^e) \quad (\text{Equation 2})$$

In Equation 2, $\text{pos}_{\text{sub.v.sup.e}}$ indicates the learnable positional embedding. Similarly, the audio encoder $E_{\text{sub.a}}$ may encode the image tiles $X_{\text{sub.a}}$ and output uni-modal features a , where:

$$(37) \quad a = \{a_{t_i}\}_{i=1}^N = E_a(X_a + \text{pos}_a^e) \quad (\text{Equation 3})$$

and $\text{pos}_{\text{sub.a.sup.e}}$ indicates the learnable positional embedding.

(38) A schematic of a process for generating sequences of audio embeddings and visual embeddings from an input video (e.g., of a process corresponding to steps **202-208** of method **200**), is illustrated in FIG. 3A. As shown, a sequence of image tiles **302** and a plurality of audio data segments **304** may be generated from an input video **300**. Image tiles **302** may be provided to a visual encoder **306**, which may output a sequence of image embeddings **310**. Audio data segments **304** may be provided to an audio encoder **308**, which may output a sequence of audio embeddings **312**.

(39) Returning to FIG. 2, after the sequence of image embeddings and the sequence of audio embeddings are generated, a first subset of the sequence of audio embeddings may be transformed into one or more synthetic image embeddings using an audio-to-visual (A2V) network (step **210** of method **200**) and a first subset of the sequence of image embeddings may be transformed into one or more synthetic audio embeddings using a visual-to-audio (V2A) network (step **212** of method **200**). The first subset of the sequence of audio embeddings and the first subset of the sequence of image embeddings may include half of the audio embeddings and half of the image embeddings, respectively. The first subset of the sequence of image embeddings may correspond to a first set of time points (e.g., temporal slices) in the input video, while the first subset of the sequence of audio embeddings may correspond to a second set of time points (e.g., temporal slices) in the input video that differ from the first set of time points.

(40) To acquire the first subset of the sequence of image embeddings and the first subset of the sequence of audio embeddings, a subset $\{\text{circumflex over } (N)\}$ of the N temporal slices may be selected. This selection may be random and may be performed using any suitable randomized selection technique. The first subset of the sequence of image embeddings may correspond to the

selected subset $\{\text{circumflex over (N)}\}$ of the N temporal slices, while the first subset of the sequence of audio embeddings may correspond to the complement $(\{\text{circumflex over (N)}\})^{\text{sup.C}}$ of the selected subset of the N temporal slices. In this case, the image embeddings that are not in the first subset of the sequence of image embeddings may belong to a second subset of the sequence of image embeddings that corresponds to the complement $(\{\text{circumflex over (N)}\})^{\text{sup.C}}$ of the selected subset of the N temporal slices, and the audio embeddings that are not in the first subset of the sequence of audio embeddings may belong to a second subset of the sequence of audio embeddings that corresponds to the selected subset $\{\text{circumflex over (N)}\}$ of the N temporal slices. Alternatively, the first subset of the sequence of image embeddings may correspond to the complement $(\{\text{circumflex over (N)}\})^{\text{sup.C}}$ of the selected subset of the N temporal slices, while the first subset of the sequence of audio embeddings may correspond to the selected subset of the N temporal slices. In this case, the image embeddings that are not in the first subset of the sequence of image embeddings may belong to a second subset of the sequence of image embeddings that corresponds to the selected subset of the N temporal slices, and the audio embeddings that are not in the first subset of the sequence of audio embeddings may belong to a second subset of the sequence of audio embeddings that corresponds to the complement $(\{\text{circumflex over (N)}\})^{\text{sup.C}}$ of the selected subset of the N temporal slices.

(41) A complementary masking process may be used to mask the image embeddings that are not in the first subset of the sequence of image embeddings and to mask the audio embeddings that are not in the first subset of the sequence of audio embeddings. If the first subset of the sequence of image embeddings corresponds to the selected subset $\{\text{circumflex over (N)}\}$ of the N temporal slices, then the image embeddings in the second subset of the sequence of image embeddings corresponding to the complement $(\{\text{circumflex over (N)}\})^{\text{sup.C}}$ of the selected subset of the N temporal slices may be masked. In this case, the first subset of the sequence of audio embeddings corresponds to the complement $(\{\text{circumflex over (N)}\})^{\text{sup.C}}$ of the selected subset of the N temporal slices, so the second subset of the sequence of audio embeddings corresponding to the selected subset $\{\text{circumflex over (N)}\}$ of the N temporal slices may be masked. Alternatively, if the first subset of the sequence of image embeddings corresponds to the complement $(\{\text{circumflex over (N)}\})^{\text{sup.C}}$ of the selected subset of the N temporal slices, then the image embeddings in the second subset of the sequence of image embeddings corresponding to the selected subset $\{\text{circumflex over (N)}\}$ of the N temporal slices may be masked. In this case, the first subset of the sequence of audio embeddings corresponds to the selected subset $\{\text{circumflex over (N)}\}$ of the N temporal slices, so the second subset of the sequence of audio embeddings corresponding to complement $(\{\text{circumflex over (N)}\})^{\text{sup.C}}$ of the selected subset of the N temporal slices may be masked.

(42) Masking of the image embeddings that are not in the first subset of the sequence of image embeddings may be performed using a visual masking module $M_{\text{sub.v}}$ and masking of the audio embeddings that are not in the first subset of the sequence of audio embeddings may be performed using an audio masking module $M_{\text{sub.a}}$. The visible image embeddings following masking (that is, the unmasked image embeddings) may make up the first subset of the sequence of image embeddings and may be given by Equation 4:

$$(43) \quad v_{\text{vis}} = M_v \odot v \quad (\text{Equation4})$$

The masked image embeddings may be the image embeddings that are not in the first subset of the sequence of image embeddings (that is, the image embeddings in the second subset of image embeddings) and may be given by Equation 5:

$$(44) \quad v_{\text{msk}} = (M_v) \odot v \quad (\text{Equation5})$$

In Equations 4-5, $\odot$ may represent the Hadamard product and $\neg$ may represent the logical NOT operator.

(45) Similarly, the visible audio embeddings following masking (that is, the unmasked audio

embeddings) may make up the first subset of the sequence of audio embeddings and may be given by Equation 6:

$$(46) \ a_{\text{vis}} = M_a \odot a \quad (\text{Equation 6})$$

The masked audio embeddings may be the audio embeddings that are not in the first subset of the sequence of audio embeddings (that is, the audio embeddings in the second subset of audio embeddings) and may be given by Equation 7:

$$(47) \ a_{\text{msk}} = (M_a) \odot a \quad (\text{Equation 7})$$

In Equations 6-7, $\odot$ may represent the Hadamard product and $\neg$ may represent the logical NOT operator.

(48) The visual mask and the audio mask may be complementary binary masks—that is, $(M.\text{sub}.a, M.\text{sub}.v) \in \{0,1\}$ such that $M.\text{sub}.a=1$ for time points where $M.\text{sub}.v=0$ and vice versa. In other words, for every masked audio embedding, the corresponding image embedding may be visible (i.e., an element of the first subset) and vice versa.

(49) A schematic of a complementary masking process is illustrated in FIG. 3B. As shown, the sequence of image embeddings **310** may be provided to a visual masking module **314**, which may mask a portion of the sequence of image embeddings **310** and output a first subset of visible image embeddings **318**. Likewise, the sequence of audio embeddings **312** may be provided to an audio masking module **316**, which may mask a portion of the sequence of audio embeddings **312** and output a first subset of visible audio embeddings **320**. The temporal slices corresponding to the visible image embeddings **318** may be complementary to the temporal slices corresponding to the visible audio embeddings **320**.

(50) Transforming the first subset of the sequence of image embeddings and the first subset of the sequence of audio embeddings (steps **210** and **212** of method **200** shown in FIG. 2) may be performed using a feature fusion process. The A2V network that transforms the first subset of the sequence of audio embeddings into one or more synthetic image embeddings may be an A2V network that has been trained to create synthetic image embeddings $v.\text{sub}.a$ that are cross-modal temporal counterparts to the first subset of the sequence of audio embeddings. Specifically, the A2V network may be trained to create cross-modal temporal counterparts $v.\text{sub}.a = A2V(a.\text{sub}.vis)$ to the unmasked/visible audio embeddings $a.\text{sub}.vis$. That is, in some embodiments,

$$(51) \ v_a = \{v_{t_i, a} = A2V(a_{t_i}) \forall t_i \text{ where } a_{t_i} \in a_{\text{vis}}\} \quad (\text{Equation 8})$$

Likewise, the V2A network that transforms the first subset of the sequence of image embeddings into one or more synthetic audio embeddings may be a trained V2A network that has been trained to create synthetic audio embeddings $a.\text{sub}.v$ that are cross-modal temporal counterparts to the first subset of the sequence of image embeddings. Specifically, the V2A network may be trained to create cross-modal temporal counterparts $a.\text{sub}.v = V2A(v.\text{sub}.vis)$ to the unmasked/visible audio embeddings $v.\text{sub}.vis$. That is, in some embodiments,

$$(52) \ a_v = \{a_{t_i, v} = V2A(v_{t_i}) \forall t_i \text{ where } v_{t_i} \in v_{\text{vis}}\} \quad (\text{Equation 9})$$

Each of the A2V and V2A networks may comprise a single layer multilayer perceptron (MLP) to match the number of tokens of the other modality followed by a single transformer block.

(53) Once the synthetic image embeddings have been generated using the A2V network, the sequence of image embeddings may be updated by replacing the second subset of the sequence of image embeddings (i.e., the subset of the image embeddings that are not in the first subset, e.g., the subset $v.\text{sub}.msk$ of the sequence of image embeddings that were masked) with the synthetic image embeddings (step **214** of method **200**). Similarly, once the synthetic audio embeddings are generated using the V2A network, the sequence of audio embeddings may be updated by replacing the second subset of the sequence of audio embeddings (i.e., the subset of the audio embeddings that are not in the first subset, e.g., the subset $a.\text{sub}.msk$ of the sequence of audio embeddings that were masked) with the synthetic audio embeddings (step **216** of method **200**).

(54) The sequences of image embeddings and audio embeddings may be updated using cross-modal fusion that replaces the second subsets of each sequence with cross-modal slices generated from the corresponding slices in the other modality. For example, the sequence of audio embeddings a may be updated by replacing each masked slice $a.sub.msk$ with the corresponding slice of the same temporal index in the cross-modal vector $a.sub.v$ given by the V2A network to form an updated sequence of audio embeddings a' , and the sequence of image embeddings v may be updated by replacing each masked slice $v.sub.msk$ with the corresponding slice of the same temporal index in the cross-modal vector $v.sub.a$ given by the A2V network to form an updated sequence of image embeddings v' .

(55) A schematic of a cross-modal fusion process (e.g., a process corresponding to steps **210-216** of method **200**) is illustrated in FIG. 3C. The subset of visible image embeddings **318** may be provided to a V2A network **322**, which may generate synthetic audio embeddings **326** corresponding to the same temporal slices as visible image embeddings **318**. Likewise, the subset of visible audio embeddings **320** may be provided to an A2V network **320**, which may generate synthetic image embeddings **328** corresponding to the same temporal slices as visible audio embeddings **320**. Synthetic audio embeddings **326** may replace the subset of masked audio embeddings (see FIG. 3B) to form an updated sequence of audio embeddings **332** and synthetic image embeddings **328** may replace the subset of masked image embeddings (see FIG. 3B) to form an updated sequence of image embeddings **330**.

(56) After the sequence of image embeddings has been updated, it may be input into a visual decoder $G.sub.v$, which may decode the updated sequence of image embeddings to produce a reconstruction $X.sub.v.sup.R$ of the sequence of image titles $X.sub.v$ (step **218** of method **200**). Similarly, after the sequence of audio embeddings has been updated, it may be input into an audio decoder $G.sub.a$, which may decode the updated sequence of audio embeddings to produce a reconstruction $X.sub.a.sup.R$ of the plurality of audio data segments $X.sub.a$ (step **220** of method **200**). The decoders may use a transformer-based architecture and may be configured to utilize the mix of uni-modal slices and cross-modal slices present in the updated sequence of image embeddings and the updated sequence of audio embeddings to generate reconstructions for the visual modality and the audio modality. In some embodiments,

$$(57) \quad 0 \quad X_v^R = G_v(v' + pos_v^g) \quad (\text{Equation 10}) \quad \text{and} \quad X_a^R = G_a(a' + pos_a^g) \quad (\text{Equation 11})$$

where $pos.sub.v.sup.g$ and $pos.sub.a.sup.g$ are the learnable positional embeddings for the visual modality and the audio modality, respectively.

(58) A schematic of a process for decoding process (e.g., a process corresponding to steps **218-220** of method **200**) is illustrated in FIG. 3D. The updated sequence of image embeddings **330** may be input into a visual decoder **334** and the updated sequence of audio embeddings **332** may be input into an audio decoder **336**. Using the updated sequence of image embeddings **330**, visual decoder **334** may produce a reconstruction **338** of the plurality of image tiles from the input video. Using the updated sequence of audio embeddings **332**, audio decoder **336** may produce a reconstruction **340** of the plurality of audio data segments from the input video.

(59) Executing method **200** for a plurality of input videos may train the audio and visual encoders, the A2V and V2A networks, and the audio and visual decoders. For the learning, a dual-objective loss function may be employed. This loss function may compute an audio-visual contrastive loss $\text{custom character.sub.C}$ between the audio and visual feature embeddings and an autoencoder loss $\text{custom character.sub.ae}$ between the input audio/visual data and the reconstructed audio/visual data.

(60) The audio-visual contrastive loss $\text{custom character.sub.C}$ may be configured to enforce similarity constraints between the audio and visual embeddings of a given input video. In some embodiments, the audio-visual contrastive loss $\text{custom character.sub.C}$ is defined as follows:

(61)

$$\mathcal{L}_C = - \sum_{p, q \in \{a, v\}, p \neq q} \frac{1}{2N} \sum_{i=1}^N \text{Math. log} \left[\frac{\exp(\text{Math. } \bar{p}^{(i)} \cdot \text{Math. } \bar{q}^{(i)})}{\sum_{j=1}^N \exp(\text{Math. } \bar{p}^{(i)} \cdot \text{Math. } \bar{q}^{(j)})} \right] \quad (\text{Equation12})$$

In Equation 12, $\text{p.sup.(i)} = \text{Mean}(\text{p.sup.(i)})$ is the mean latent (embedding) vector across the patch dimension of the uni-modal embeddings of the i-th data sample, N is the number of video samples, τ is a temperature parameter that controls the spread of the distribution

$$(62) \quad \frac{\exp(\text{Math. } \bar{p}^{(i)} \cdot \text{Math. } \bar{q}^{(i)})}{\sum_{j=1}^N \exp(\text{Math. } \bar{p}^{(i)} \cdot \text{Math. } \bar{q}^{(j)})}$$

and i, j are sample indices.

(63) The autoencoder loss  may be composed of a reconstruction loss  and an adversarial loss . The reconstruction loss  may be computed between the plurality of image tiles and the plurality of audio data segments (X.sub.v, X.sub.a) and their respective reconstructions (X.sub.v.sup.R, X.sub.a.sup.R) and may be computed only over the masked tokens. In some embodiments, the reconstruction loss  is defined by equation 13:

$$(64) \quad \mathcal{L}_{\text{rec}} = \sum_{p \in \{a, v\}} \frac{1}{N} \sum_{i=1}^N \text{Math. } M_p \cdot \text{Math. } X_p^{(i)} - M_p \cdot \text{Math. } X_p^R \quad (\text{Equation13})$$

(65) The adversarial loss  may be configured to supplement the reconstruction loss by enhancing the features captured in the reconstructions of each modality. Similar to the reconstruction loss, the adversarial loss may be computed only on the masked tokens. In some embodiments, the Wasserstein GAN loss is used for the adversarial loss.

(66) Training with an adversarial loss can comprise a generator training step and a discriminator training step. During the generator training step, the computed loss is back propagated through the entire model pipeline, including the encoder, A2V network, V2A network, and decoder. During the discriminator training step, the loss is back propagated through a separate discriminator network comprising a multilayer perceptron (MLP). In some embodiments, the adversarial loss during the generator and the discriminator training steps ( and , respectively) are given by equations 14-15:

$$(67) \quad \mathcal{L}_{\text{adv}}^{(G)} = - \sum_{p \in \{a, v\}} \frac{1}{N} \sum_{i=1}^N \text{Math. } D_p(M_p \cdot \text{Math. } X_p^R) \quad (\text{Equation14})$$

$$\mathcal{L}_{\text{adv}}^{(D)} = - \sum_{p \in \{a, v\}} \frac{1}{N} \sum_{i=1}^N \text{Math. } (D_p(M_p \cdot \text{Math. } X_p^R) - D_p(M_p \cdot \text{Math. } X_p^{(i)})) \quad (\text{Equation15})$$

In Equations 14-15, D.sub.p denotes the discriminator of each modality.

(68) The overall training loss  for the generative training step and the overall training loss  for the discriminative training step may be given by:

$$(69) \quad \mathcal{L}^{(G)} = \lambda_c \mathcal{L}_C + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}^{(G)} \quad (\text{Equation16}) \quad \mathcal{L}^{(D)} = \mathcal{L}_{\text{adv}}^{(D)} \quad (\text{Equation17})$$

where $\lambda_{\text{sub.c}}$, $\lambda_{\text{sub.rec}}$, and $\lambda_{\text{sub.adv}}$ represent loss weights for the contrastive loss, the reconstruction loss, and the adversarial loss, respectively. Computing the autoencoding loss on the masked temporal slices may strictly enforce the decoders of each modality to learn from the other modality, as the input embeddings for the decoder at masked indices are obtained from the other modality. This strategy may explicitly enforce audio-visual correspondence supplementing the contrastive loss objective.

(70) As noted above, to train the audio and visual encoders, the A2V and V2A networks, and the audio and visual decoders, method 200 may be executed for a plurality of input videos. These input videos may be real videos (e.g., videos that are not AI-generated). In some embodiments, the input videos may show human faces. Working exclusively with real face videos during training may cause the model to learn the dependency between “real” speech audio and the corresponding visual

facial features.

(71) After the audio encoder, the visual encoder, the A2V network, and the V2A network have been trained (e.g., by executing method **200** for a plurality of real input videos), the trained encoders and cross-modal networks may be used to train a classifier to detect fake videos. FIG. **4** provides an exemplary method **400** for training a classifier to detect fake videos. Method **400** may be executed using a computer system and may produce a trained classifier such as classifier **106** shown in FIG. **1**. The computer system used to perform method **400** may, for example, comprise one or more electronic devices implementing a software platform. In other examples, the computer system may be a client-server system, in which case the blocks of method **400** may be divided up between the server and a client device, between the server and multiple client devices, using only a client device, or using only multiple client devices.

(72) In various embodiments of method **400**, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the blocks of method **400**. Accordingly, the operations as illustrated (and described in greater detail below) are exemplary by nature and, as such, should not be viewed as limiting.

(73) A preprocessing stage may be performed prior to executing method **400**. In the preprocessing stage, an input video may be processed to extract image data and audio data using the same techniques described above with respect to method **200**. For method **400**, the input video may be a sample from a labeled dataset $D_{\text{sub.f}}$ comprising both real and fake videos—that is, the input video may be a sample $(x, y) \in D_{\text{sub.f}}$, where x is the video and y is a label indicating whether the video is real or fake. A sequence of image tiles $X_{\text{sub.v}}$ may then be generated from the image data that is extracted from the input video (step **402** of method **400**) and a sequence of audio data segments $X_{\text{sub.a}}$ may be generated from the audio data that is extracted from the input video (step **404** of method **400**). The sequence of image tiles and the plurality of audio data segments may be generated using the techniques discussed above with respect to steps **202-204** of method **200**.

(74) After the sequence of image tiles is generated, it may be provided as input to a trained visual encoder $E_{\text{sub.v}}$ to generate a sequence of image embeddings v (step **406** of method **400**), e.g., as defined by Equation 2. Likewise, after the plurality of audio data segments is generated, it may be provided as input to a trained audio encoder $E_{\text{sub.a}}$ to generate a sequence of audio embeddings a (step **408** of method **400**), e.g., as defined by Equation 3. The trained visual encoder and the trained audio encoder may have been trained using a protocol such as method **200** (FIG. **2**).

(75) The sequence of image embeddings v may be provided as input to a trained visual-to-audio (V2A) network to obtain a sequence of synthetic audio embeddings $a_{\text{sub.v}}$ (step **410** of method **400**). The sequence of synthetic audio embeddings $a_{\text{sub.v}}$ can be defined as follows:

$$(76) \quad a_v = \{a_{t_i, v} = V2A(v_{t_i}) \forall t_i \text{ where } v_{t_i} \in v\} \quad (\text{Equation 18})$$

(77) Similarly, the sequence of audio embeddings a may be provided as input to a trained audio-to-visual (A2V) network to obtain a sequence of synthetic image embeddings $v_{\text{sub.a}}$ (step **412** of method **400**). The sequence of synthetic image embeddings $v_{\text{sub.a}}$ can be defined as follows:

$$(78) \quad v_a = \{v_{t_i, a} = A2V(a_{t_i}) \forall t_i \text{ where } a_{t_i} \in a\} \quad (\text{Equation 19})$$

The trained V2A network and the trained A2V network may have been trained using a protocol such as method **200**.

(79) After the sequence of synthetic audio embeddings has been obtained, it may be concatenated with the sequence of audio embeddings that was generated by the audio encoder to produce a combined audio embedding sequence $f_{\text{sub.a}}$ (step **414** of method **400**), where:

$$(80) \quad f_a = a \oplus a_v \quad (\text{Equation 20})$$

The sequence of synthetic image embeddings may likewise be concatenated with the sequence of image embeddings that was generated by the visual encoder to produce a combined image embedding sequence $f_{\text{sub.v}}$ (step **414** of method **400**), where:

$$(81) f_v = v \oplus v_a \quad (\text{Equation 21})$$

In Equations 20-21, $\oplus$ represents the concatenation operator along the feature dimension.

(82) The combined audio embedding sequence and the combined image embedding sequence may then be provided as input to a classifier, which may determine whether the input video is real or fake (step **416** of method **400**). In some embodiments, the classifier is a classifier network Q comprising two uni-modal patch reduction networks: an audio mode patch reduction network $\Psi_{\text{sub.a}}$ and a visual mode patch reduction network $\Psi_{\text{sub.v}}$. The patch reduction networks may be followed by a classifier head Γ . Each combined embedding sequence, $f_{\text{sub.a}}$, $f_{\text{sub.v}}$ may first be distilled in the patch dimension using the corresponding uni-modal patch reduction networks. The output embeddings may then be concatenated along the feature dimension and fed into the classifier head. The classifier head may output the logits l used to classify if a given sample is real or fake. In some embodiments,

$$(83) l = Q(f_a, f_v) = (\Psi_{\text{sub.a}}(f_a) \oplus \Psi_{\text{sub.v}}(f_v)) \quad (\text{Equation 22})$$

A cross-entropy loss  custom character.sub.CE may be used as the learning objective and may be computed using the label y on the input video that indicates whether the input video is real or fake and the output logits l .

(84) During inference, a video may be split into blocks of time T (the sample length during training) with a step size of T/N , which is the duration of a temporal slice. The output logits can be computed for each of the blocks and the classification decision (real or fake) can be made based on the mean of the output logits.

(85) A schematic of a process for training a classifier to detect fake videos (corresponding to method **400** shown in FIG. 4) is illustrated in FIGS. 5A-5D. A sequence of image tiles **502** and a plurality of audio data segments **504** may be generated from a labeled input video **500**. Image tiles **502** may be provided to a trained visual encoder **506**, which may output a sequence of image embeddings **510**. Audio data segments **504** may be provided to a trained audio encoder **508**, which may output a sequence of audio embeddings **512** (FIG. 5A). Image embeddings **510** may be provided as input to a trained V2A network **514** to generate a sequence of synthetic audio embeddings **518**. Audio embeddings **512** may be provided as input to a trained A2V network **516** to generate a sequence of synthetic image embeddings **520** (FIG. 5B). Image embeddings **510** and synthetic image embeddings **520** may be concatenated to form a combined sequence of image embeddings **522**. Audio embeddings **512** and synthetic audio embeddings **518** may be concatenated to form a combined sequence of audio embeddings **524** (FIG. 5C). The combined sequence of image embeddings **522** and the combined sequence of audio embeddings **524** may then be provided as input to a classifier **526**. Classifier **526** may output logits **528** that can be used to classify whether input video **500** is real or fake. To train classifier **526**, a cross-entropy loss may be computed using logits **528** and the label of input video **500**.

(86) FIG. 6 shows an exemplary computer system **600** that can be used to execute the described methods (e.g., method **200**, method **400**). Computer system **600** can be any suitable type of microprocessor-based device, such as a personal computer, workstation, server, or handheld computing device (portable electronic device) such as a phone or tablet, or dedicated device. As shown in FIG. 6, computer system **600** may include one or more classical (binary) processors **602**, an input device **604**, an output device **606**, storage **608**, and a communication device **612**.

(87) Input device **604** and output device **606** can be connectable or integrated with system **102**. Input device **604** may be any suitable device that provides input, such as a touch screen, keyboard or keypad, mouse, or voice-recognition device. Likewise, output device **606** can be any suitable device that provides output, such as a display, touch screen, haptics device, or speaker.

(88) Storage **608** can be any suitable device that provides storage, such as an electrical, magnetic, or optical memory, including a RAM, cache, hard drive, removable storage disk, or other non-transitory computer readable medium. Communication device **612** can include any suitable device

capable of transmitting and receiving signals over a network, such as a network interface chip or device. The components of computer system **600** can be connected in any suitable manner, such as via a physical bus or via a wireless network.

(89) Processor(s) **602** may be or comprise any suitable classical processor or combination of classical processors, including any of, or any combination of, a central processing unit (CPU), a field programmable gate array (FPGA), and an application-specific integrated circuit (ASIC). Software **610**, which can be stored in storage **608** and executed by processor(s) **602**, can include, for example, the programming that embodies the functionality of the present disclosure. Software **610** may be stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a computer-readable storage medium can be any medium, such as storage **608**, that can contain or store programming for use by or in connection with an instruction execution system, apparatus, or device.

(90) Software **610** can also be propagated within any transport medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a transport medium can be any medium that can communicate, propagate, or transport programming for use by or in connection with an instruction execution system, apparatus, or device. The transport readable medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic, or infrared wired or wireless propagation medium.

(91) Computer system **600** may be connected to a network, which can be any suitable type of interconnected communication system. The network can implement any suitable communications protocol and can be secured by any suitable security protocol. The network can comprise network links of any suitable arrangement that can implement the transmission and reception of network signals, such as wireless network connections, T1 or T3 lines, cable networks, DSL, or telephone lines.

(92) Computer system **600** can implement any operating system suitable for operating on the network. Software **610** can be written in any suitable programming language, such as C, C++, Java, or Python. In various embodiments, application software embodying the functionality of the present disclosure can be deployed in different configurations, such as in a client/server arrangement or through a Web browser as a Web-based application or Web service, for example.

EXAMPLES

Example 1

(93) Visual and audio encoders and audio-to-visual (A2V) and visual-to-audio (V2A) networks are trained using the LRS3 dataset (Triantafyllos Afouras, Joon Son Chung, and Andrew Zisserman. Lrs3-ted: a large-scale dataset for visual speech recognition. arXiv preprint arXiv:1809.00496, 2018), which exclusively contains real videos. The trained encoders and A2V and V2A networks are then used to train classifier following a supervised learning approach using the FakeAVCeleb dataset (Hasam Khalid, Shahroz Tariq, Minha Kim, and Simon S Woo. Fakeavceleb: A novel audio-video multimodal deepfake dataset. arXiv preprint arXiv:2108.05080, 2021). FakeAVCeleb comprises both real and fake videos, where either one or both audio-visual modalities have been synthesized using different combinations of several generative deepfake algorithms (visual: FaceSwap (Iryna Korshunova, Wenzhe Shi, Joni Dambre, and Lucas Theis. *Fast face-swap using convolutional neural networks*. In *Proceedings of the IEEE international conference on computer vision*, pages 3677-3685, 2017), FSGAN (Yuval Nirkin, Yosi Keller, and Tal Hassner. Fsgan: Subject agnostic face swapping and reenactment. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7184-7193, 2019), and Wav2Lip (K R Prajwal, Rudrabha Mukhopadhyay, Vinay P Namboodiri, and CV Jawahar. A lip sync expert is all you need for speech

to lip generation in the wild. In *Proceedings of the 28th ACM international conference on multimedia*, pages 484-492, 2020); audio: SV2TTS (Ye Jia, Yu Zhang, Ron Weiss, Quan Wang, Jonathan Shen, Fei Ren, Patrick Nguyen, Ruoming Pang, Ignacio Lopez Moreno, Yonghui Wu, et al. Transfer learning from speaker verification to multispeaker text-to-speech synthesis. *Advances in neural information processing systems*, 31, 2018)).

(94) Dataset Details

(95) LRS3: This dataset introduced by Afouras et al. exclusively comprises of real videos. It consists of 5594 videos spanning over 400 hours of TED and TED-X talks in English. The videos in the dataset are processed such that each frame contains faces and the audio and visual streams are in sync.

(96) FakeAVCeleb: The FakeAVCeleb dataset is a deepfake detection dataset, which consists of 20,000 video clips in total. It comprises of 500 real videos sampled from the VoxCeleb2 and 19500 deepfake samples generated using different manipulation methods applied on the set of real videos. The dataset consists of the following manipulations, where the deepfake algorithms used in each category are indicated within brackets: RVFA: Real Visuals-Fake Audio (SV2TTS) FVRA-FS: Fake Visuals-Real Audio (FaceSwap) FVFA-FS: Fake Visuals-Fake Audio (SV2TTS+FaceSwap) FVFA-GAN: Fake Visuals-Fake Audio (SV2TTS+FaceSwapGAN) FVRA-GAN: Fake Visuals-Real Audio (FaceSwapGAN) FVRA-WL: Fake Visuals-Real Audio (Wav2Lip) FVFA-WL: Fake Visuals-Fake Audio (SV2TTS+Wav2Lip)

(97) KoDF: This dataset is a large-scale dataset comprising real and synthetic videos of 400+ subjects speaking Korean. KoDF consists of 62K+ real videos and 175K+ fake videos synthesized using the following six algorithms: FaceSwap, DeepFaceLab, FaceSwapGAN, FOMM, ATFHP, and Wav2Lip. A subset of this dataset is used to evaluate the cross-dataset generalization performance of the model.

(98) DFDC: The DeepFake Detection Challenge (DFDC) dataset is another deepfake dataset that consists of samples with fake audio besides FakeAVCeleb. It consists of over 100K video clips in total generated using deepfake algorithms such as MM/NN Face Swap, NTH, FaceSwapGAN, StyleGAN, and TTS Skins. A subset of this dataset consisting of 3215 videos is used to evaluate the model's cross-dataset generalization performance.

(99) DF-TIMIT: The Deepfake TIMIT dataset comprises deepfake videos manipulated using FaceSwapGAN. The real videos used for manipulation have been sourced by sampling similar looking identities from the VidTIMIT dataset. The higher-quality (HQ) version, which consists of 320 videos, was used in evaluating cross-dataset generalization performance.

(100) Representation Learning Stage

(101) Samples are drawn from the LRS3 dataset, which exclusively contains real videos. The audio stream is converted to a Mel-spectrogram of 128 Mel-frequency bins, with a 16 ms Hamming window every 4 ms. Video clips of $T=3.2$ s in duration are randomly sampled, sampling 16 visual frames and 768 audio frames (Mel) with clipping/padding where necessary. The 16 visual frames are uniformly sampled such that they are at the first and third quartile of a temporal slice (2 frames/slice $\times$ 8 slices). The visual frames are resized to 224×224 spatially and are augmented using random grayscaling and horizontal flipping, each with a probability of 0.5. We make sure that in a given batch, for each sample we draw another sample from the same video but at a different time interval to make sure the model is exposed to the notion of temporal shifts when computing the contrastive loss. Both audio and visual modalities are normalized.

(102) The encoder and decoder architectures of each modality are adopted from the VideoMAE (Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Video-mae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. *Advances in neural information processing systems*, 35:10078-10093, 2022) based on ViT-B (Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16×16 words: Transformers for

image recognition at scale. In International Conference on Learning Representations, 2020). Each of the A2V/V2A networks is composed of a linear layer to match the number of tokens of the other modality followed by a single transformer block.

(103) The audio encoder and decoder are initialized using the checkpoint of AudioMAE (Po-Yao Huang, Hu Xu, Juncheng Li, Alexei Baevski, Michael Auli, Wojciech Galuba, Florian Metze, and Christoph Feichtenhofer. Masked autoencoders that listen. *Advances in Neural Information Processing Systems*, 35:28708-28720, 2022) pretrained on AudioSet-2M (Jort F Gemmeke, Daniel P W Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R Channing Moore, Manoj Plakal, and Marvin Ritter. Audio set: An ontology and human-labeled dataset for audio events. In 2017 IEEE international conference on acoustics, speech and signal processing (ICASSP), 219 pages 776-780. IEEE, 2017) and the visual encoder and decoder using the checkpoint of MARLIN (Zhixi Cai, Shreya Ghosh, Kalin Stefanov, Abhinav Dhall, Jianfei Cai, Hamid Rezaatofighi, Reza Haffari, and Munawar Hayat. Marlin: Masked autoencoder for facial video representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1493-1504, 2023) pretrained on the YouTubeFace (Lior Wolf, Tal Hassner, and Itay Maoz. Face recognition in unconstrained videos with matched background similarity. In *CVPR 2011*, pages 529-534. IEEE, 2011) dataset. Subsequently, the representation learning framework is trained end-to-end using the AdamW optimizer (Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2018) with a learning rate of $1.5e-4$ with a cosine decay. The weights of the losses are as follows: $\lambda_{\text{sub.c}}=0.01$, $\lambda_{\text{sub.rec}}=1.0$, and $\lambda_{\text{sub.adv}}=0.1$, which were chosen empirically. Training is performed for 500 epochs with a linear warmup for 40 epochs using a batch size of 32 and a gradient accumulation interval of 2. The training was performed on 4 RTX A6000 GPUs for approximately 60 hours.

(104) Deepfake Classification Stage

(105) Samples are drawn from FakeAVCeleb, which consists of deepfake videos where either or both audio and visual modalities have been manipulated. The preprocessing and sampling strategy is similar to that of representation learning stage, except an additional sample is not drawn from the same video clip as a contrastive learning objective is not used at this stage. Weighted sampling is employed to mitigate the issue of class imbalance between real and fake samples.

(106) Each of the uni-modal patch reduction networks is a 3-layer MLP, while the classifier head is a 4-layer MLP. No changes are made to the representation learning architecture.

(107) The representation learning framework is initialized using the pretrained checkpoint obtained from the representation learning stage. Subsequently, the pipeline is trained end-to-end using the AdamW optimizer with a cosine annealing with warm restarts scheduler with a maximum learning rate of $1.0e-4$ for 50 epochs with a batch size of 32. The training was performed on 4 RTX A6000 GPUs for approximately 10 hours.

(108) Evaluation

(109) The performance of the model is evaluated against the existing state-of-the-art algorithms on multiple criteria: intra-dataset performance, cross-manipulation performance, and cross-dataset generalization. The results are compared against both uni-modal (visual) state-of-the-art approaches and audio-visual approaches based on accuracy (ACC), average precision (AP), and area under the ROC curve (AUC). The average results across multiple runs with different random seeds are reported. Further, for audio-visual algorithms, a video is labeled as fake if either or both audio and visual modalities have been manipulated. For uni-modal algorithms, a video is considered fake only if the visual modality has been manipulated to maintain fairness.

(110) The model training utilizes 70% of all FakeAVCeleb samples, while the remaining 30% constitutes the unseen test set. Table 1 summarizes the performance of the model (titled “AVFF” in Table 1) against baselines using a 70-30 train-test split on the FakeAVCeleb dataset. As denoted in Table 1, the AVFF approach demonstrates substantial improvements over the existing state-of-the-art, both in audio-visual (AVoiD-DF) and uni-modal (RealForensics) deepfake detection. Compared

to AVoiD-DF, the AVFF model achieves an increase in accuracy of 14.9% (+9.9% in AUC) and compared to RealForensics the accuracy increases by 8.7% (+4.5% AUC). Overall, the superior performance of audio-visual methods leveraging cross-modal correspondence is evident, outperforming uni-modal approaches that rely on uni-modal artifacts (i.e., visual anomalies) introduced by deepfake algorithms. RealForensics, while competitive, discards the audio modality during detection, limiting its applicability exclusively to visual deepfakes. This hinders its practicality as contemporary deepfakes often involve manipulations in both audio and visual modalities. The enhanced results of both RealForensics and our proposed method highlight the positive impact of employing a pre-training stage for effective representation learning.

(111) TABLE-US-00001 TABLE 1 Intra-dataset performance Method Modality ACC AUC
Xception V 67.9 70.5 LipForensics V 80.1 82.4 FTCN [51] V 64.9 84.0 CViT [45] V 69.7 71.8
RealForensics [18] V 89.9 94.6 Emotions Don't Lie AV 78.1 79.8 [34] MDS [7] AV 82.8 86.5
AVFakeNet [24] AV 78.4 83.4 VFD [6] AV 81.5 86.1 AVoiD-DF [48] AV 83.7 89.2 AVFF AV 98.6
99.1

(112) Additional information about the algorithms against which the model was compared can be found in the following references: Xception: Andreas Rossler, Davide Cozzolino, Luisa Verdoliva, Christian Riess, Justus Thies, and Matthias Nießner. Faceforensics++: Learning to detect manipulated facial images. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 1-11, 2019. LipForensics: Alexandros Haliassos, Konstantinos Vougioukas, Stavros Petridis, and Maja Pantic. Lips don't lie: A generalisable and robust approach to face forgery detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5039-5049, 2021. FTCN: Yinglin Zheng, Jianmin Bao, Dong Chen, Ming Zeng, and Fang Wen. Exploring temporal coherence for more general video face forgery detection. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 15044-15054, 2021. CViT: Deressa Wodajo and Solomon Atnafu. Deepfake video detection using convolutional vision transformer. *arXiv preprint arXiv: 2102.11126*, 2021. RealForensics: Alexandros Haliassos, Rodrigo Mira, Stavros Petridis, and Maja Pantic. Leveraging real talking faces via self-supervision for robust forgery detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14950-14962, 2022. Emotions Don't Lie: Trisha Mittal, Uttaran Bhattacharya, Rohan Chandra, Aniket Bera, and Dinesh Manocha. Emotions don't lie: An audio-visual deepfake detection method using affective cues. In *Proceedings of the 28th ACM international conference on multimedia*, pages 2823-2832, 2020. MDS: Komal Chugh, Parul Gupta, Abhinav Dhall, and Ramanathan Subramanian. Not made for each other-audiovisual dissonance-based deepfake detection and localization. In *Proceedings of the 28th ACM international conference on multimedia*, pages 439-447, 2020. AVFakeNet: Hafsa Ilyas, Ali Javed, and Khalid Mahmood Malik. Avfakenet: A unified end-to-end dense swin transformer deep learning model for audio-visual deepfakes detection. *Applied Soft Computing*, 136:110124, 2023. VFD: Harry Cheng, Yangyang Guo, Tianyi Wang, Qi Li, Xiaojun Chang, and Liqiang Nie. Voice-face homogeneity tells deepfake. *arXiv preprint arXiv: 2203.02195*, 2022. AVoiD-DF: Wenyan Yang, Xiaoyu Zhou, Zhikai Chen, Bofei Guo, Zhongjie Ba, Zhihua Xia, Xiaochun Cao, and Kui Ren. Avoid-df: Audio-visual joint learning for detecting deepfake. *IEEE Transactions on Information Forensics and Security*, 18:2015-2029, 2023.

(113) The model's performance is assessed on samples generated using previously unseen manipulation methods. The scalability of deepfake detection algorithms to unseen manipulation methods is crucial for adapting to evolving threats, thus ensuring wide applicability across diverse scenarios. The FakeAVCeleb is partitioned dataset into five categories—(i) RVFA: Real Visual-Fake Audio (SV2TTS), (ii) FVRA-WL: Fake Visual-Real Audio (Wav2Lip), (iii) FVFA-FS: Fake Visual-Fake Audio (FaceSwap+Wav2Lip+SV2TTS), (iv) FVFA-GAN: Fake Visual-Fake Audio (FaceSwapGAN+Wav2Lip+SV2TTS), and (v) FVFA-WL: Fake Visual-Fake Audio (Wav2Lip+SV2TTS)-based on the algorithms used to generate the deepfakes. The model is

evaluated using these categories, leaving one category out for testing while training on the remaining categories. Results are reported in Table 2. The AVFF achieves the best performance in almost all cases (and at par with the rest) and, notably, yields consistently enhanced performance (AUC>48.4 92+%, AP>93+%) across all categories, while other baselines (Xception, LipForensics, FTCN, AVDFD) fall short in categories FVFA-GAN and RVFA.

(114) TABLE-US-00002 TABLE 2 FakeAVCeleb Cross-Manipulation Generalization FVFA-FVFA- FVFA- RVFA WL FVFA-FS GAN WL AVG-FV Method Modality AP AUC AP AUC AP AUC AP AUC AP AUC Xception V — — 88.2 88.3 92.3 93.5 67.6 68.5 91.0 91.0 84.8 85.3 LipForensics V — — 97.8 99/100 99/100 61.5 68.1 98.6 98.7 89.4 91.1 FTCN V — — 96.2 97.4 77.4 78.3 95.6 96.5 92.3 93.1 RealForensics V — — 88.8 93.0 99.3 99.1 99.8 99.8 93.4 96.7 95.3 97.1 AV-DFD AV 74.9 73.3 97.0 97.4 99.6 99.7 58.4 55.4 100. 100. 88.8 88.1 AVAD (LRS2) AV 62.4 71.6 93.6 93.7 95.3 95.8 94.1 94.3 93.8 94.1 94.2 94.5 AVAD (LRS3) AV 70.7 80.5 91.1 93.0 91.0 92.3 91.6 92.7 91.4 93.1 91.3 92.8 AVFF AV 93.3 92.4 94.8 98.2 100. 100. 99.9 100. 99.4 99.8 98.5 99.5

(115) The adaptability of the model to a different data distribution is evaluated by testing on a subset of the KoDF dataset, as well as on the DF-TIMIT dataset and a subset of the DFDC dataset (Tables 3A-3B).

(116) TABLE-US-00003 TABLE 3A Cross-dataset generalization on KoDF dataset (AP: Average Precision; AUC: Area under ROC curve) Method Modality AP AUC Xception V 76.9 77.7 LipForensics V 89.5 86.6 FTCN V 66.8 68.1 RealForensics V 95.7 93.6 AV-DFD AV 79.6 82.1 AVAD AV 87.6 86.9 AVFF AV 93.1 95.5

(117) TABLE-US-00004 TABLE 3B Cross-dataset generalization on DF-TIMIT and DFDC datasets (AP: Average Precision; AUC: Area under ROC curve) DF-TIMIT DFDC Method Modality AP AUC AP AUC Xception V 86.0 90.5 68.0 67.9 LipForensics V 96.7 98.4 76.8 77.4 FTCN V 100 99.8 70.5 71.1 RealForensics V 99.2 99.5 82.9 83.7 AVFF AV 100 100 97.0 86.2

(118) FIG. 7 shows t-distributed Stochastic Neighbor Embedding (t-SNE) plots of embeddings for random samples for each category (real videos, videos with real audio and fake visuals, videos with fake audio and real visuals, etc.) of the FakeAVCeleb dataset. As shown, distinct clusters are evident for each deepfake category, indicating that representations generated by the AVFF model are capable of capturing subtle cues that differentiate different deepfake algorithms despite not encountering any of them during the representation learning training stage. A further analysis of the t-SNE visualizations reveals that the samples belonging to adjacent clusters are related in terms of the deepfake algorithms used to generate them. For instance, FVRA-WL and FVFA-WL, which are adjacent, both employ Wav2Lip to synthesize the deepfakes (refer to the encircled regions in FIG. 7). These findings underscore the efficacy of the audio-visual representation learning paradigm.

Example 2

(119) Impact of the Autoencoding Objective

(120) In this experiment, the model is using only the contrastive loss objective, discarding the autoencoding objective, which effectively scraps away the complementary masking, cross-modality fusion, and decoding modules. The feature embeddings at the output of the encoders a, v are used for the downstream training. Results (see row (i) in Table 4) indicate a performance reduction, highlighting the importance of the autoencoding objective.

(121) Impact of the Cross-Modal Fusion

(122) In this ablation, the A2V/V2A networks, which predict the masked tokens of the other modality, are discarded, and shared learnable masked tokens similar to MAE approaches are used. The performance of the model diminishes (especially AP) (see row (ii) in Table 4). This signifies the importance of the cross-modal fusion module, as it supplements the representation of a given modality with information extracted from the other modality, which helps build the correspondence between the two modalities.

(123) Impact of Complementary Masking

(124) Replacing complementary masking with random masking results in a notable drop in AP and AUC scores, affecting the model's ability to learn correspondences (see row (iii) in Table 4). This performance drop can be attributed to the inability of the model to learn correspondences between audio and visual modalities due to the randomness, which indicates the importance of complementary masking in the proposed method.

(125) Impact of Concatenation of Different Embeddings

(126) In the deepfake classification stage, the feature embeddings (a, v) are concatenated with the cross-modal embeddings (a.sub.v, v.sub.a), creating the concatenated embeddings (f.sub.a, f.sub.v). In this experiment, the model performance is evaluated using each of the embeddings in isolation (see rows (iv) and (v) in Table 4). While the use of each embedding generates promising results, the synergy of the two embeddings enhances the performance.

(127) Impact of Uni-Modal Patch Reduction

(128) Replacing the uni-modal patch reduction networks ($\Psi_{\text{sub.a}}$, $\Psi_{\text{sub.v}}$) with Mean Pooling dents the performance slightly (see row (vi) in Table 4), which could be due to the suppression of subtle discriminative cues existing in fewer patches. Thus, the use of an MLP to reduce the patch dimension is justified, which effectively computes a weighted mean with learnable weights.

(129) TABLE-US-00005 TABLE 4 Evaluations on Ablations Method AP AUC (i) Only contrastive loss 84.2 90.3 (ii) Ours w/o cross-modality fusion 87.2 93.1 (iii) Ours w/o complementary masking 78.9 90.7 (iv) Only feature embeddings 89.7 97.6 (v) Only cross-modal embeddings 94.6 98.0 (vi) Mean Pooling features 96.5 98.1 AVFF 96.7 99.1

Example 3

(130) The performance of the model was evaluated on several unseen perturbations applied to each modality. Such perturbations may occur during video post-processing, e.g., when sharing videos through social medial platforms.

(131) Visual Perturbations

(132) The performance of the model was evaluated on the following visual perturbations: saturation, contrast, blockwise distortion, Gaussian noise, Gaussian blur, JPEG compression, and video compression on five different levels of intensities. The implementations for the perturbations and the levels of intensities were sourced from the official repository of DeeperForensics-1.0. The model's performance was compared against RealForensics. As depicted in FIGS. 8A-8B, the model demonstrated enhanced robustness against unseen visual perturbations compared to RealForensics in most scenarios. Particularly noteworthy improvements were observed in cases of block-wise distortion, Gaussian noise, and video compression.

(133) Audio Perturbations

(134) The performance of the model was evaluated on the following audio perturbations: Gaussian Noise, pitch shift, changes in reverberance, and audio compression. FIG. 9 illustrates the model's performance under these perturbations across five intensity levels. As shown, the model is robust to various audio perturbations. Notably, the model showcases high robustness to changes in reverberance, with minimal fluctuations across all intensity levels.

(135) The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments and/or examples. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

(136) As used herein, the singular forms “a”, “an”, and “the” include the plural reference unless the context clearly dictates otherwise. Reference to “about” a value or parameter or “approximately” a value or parameter herein includes (and describes) variations that are directed to that value or parameter per se. For example, description referring to “about X” includes description of “X”. It is

understood that aspects and variations of the invention described herein include “consisting of” and/or “consisting essentially of” aspects and variations.

(137) When a range of values or values is provided, it is to be understood that each intervening value between the upper and lower limit of that range, and any other stated or intervening value in that stated range, is encompassed within the scope of the present disclosure. Where the stated range includes upper or lower limits, ranges excluding either of those included limits are also included in the present disclosure.

(138) Although the disclosure and examples have been fully described with reference to the accompanying figures, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims. Finally, the entire disclosure of the patents and publications referred to in this application are hereby incorporated herein by reference.

(139) Any of the systems, methods, techniques, and/or features disclosed herein may be combined, in whole or in part, with any other systems, methods, techniques, and/or features disclosed herein.

Claims

1. A method for training a model comprising a visual encoder, an audio encoder, an audio-to-visual (A2V) network, a visual-to-audio (V2A) network, and a classifier for classifying videos as real or fake, the method comprising: generating a sequence of image tiles from image data from an input video; generating a plurality of data segments representing audio data from the input video; generating a sequence of image embeddings based on the sequence of image tiles using the visual encoder; generating a sequence of audio embeddings based on the sequence of data segments using the audio encoder; transforming, using the V2A network, a first subset of the sequence of image embeddings into one or more synthetic audio embeddings, wherein the first subset of the sequence of image embeddings corresponds to a first set of time points in the input video; transforming, using the A2V network, a first subset of the sequence of audio embeddings into one or more synthetic image embeddings, wherein the first subset of the sequence of audio embeddings corresponds to a second set of time points in the input video complementary to the first set of time points; updating the sequence of image embeddings by replacing a second subset of the sequence of image embeddings with the one or more synthetic image embeddings, wherein the second subset of the sequence of image embeddings corresponds to the second set of time points; updating the sequence of audio embeddings by replacing a second subset of the sequence of audio embeddings with the one or more synthetic audio embeddings, wherein the second subset of the sequence of audio embeddings corresponds to the first set of time points; training the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated sequence of audio embeddings; training the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network, wherein the classifier is configured to receive image embeddings for the videos from the trained visual encoder, audio embeddings for the videos from the trained audio encoder, synthetic image embeddings for the videos from the trained A2V network, and synthetic audio embeddings for the videos from the trained V2A network.

2. The method of claim 1, wherein: the first subset of the sequence of image embeddings comprises half of the image embeddings, and the first subset of the sequence of audio embeddings comprises half of the audio embeddings.

3. The method of claim 1, wherein the first subset of the sequence of image embeddings and the first subset of the sequence of audio embeddings are randomly selected.

4. The method of claim 1, wherein training the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated

- sequence of audio embeddings comprises: decoding the updated sequence of image embeddings to produce a reconstruction of the sequence of image tiles; and decoding the updated sequence of audio embeddings to produce a reconstruction of the plurality of data segments.
5. The method of claim 4, wherein the updated sequence of image embeddings is decoded using a visual decoder.
6. The method of claim 4, wherein the updated sequence of audio embeddings is decoded using an audio decoder.
7. The method of claim 1, wherein training the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated sequence of audio embeddings further comprises computing a dual-objective loss, wherein a first objective of the dual-objective loss depends on the sequence of audio embeddings and the sequence of image embeddings and a second objective of the dual-objective loss depends on the sequence of image tiles, the plurality of data segments, the reconstruction of the sequence of image tiles, and the reconstruction of the plurality of data segments.
8. The method of claim 1, wherein training the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network comprises: generating a second sequence of image tiles from image data from a labeled training video comprising a label indicating whether the labeled training video is real or fake; generating a second plurality of data segments representing audio data from the labeled training video; generating a second sequence of image embeddings based on the second sequence of image tiles using the trained visual encoder; generating a second sequence of audio embeddings based on the second sequence of data segments using the trained audio encoder; transforming, using the trained V2A network, the second sequence of image embeddings into a sequence of synthetic audio embeddings; transforming, using the trained A2V network, the second sequence of audio embeddings into a sequence synthetic image embeddings; concatenating the second sequence of image embeddings and the sequence of synthetic image embeddings to produce a combined sequence of image embeddings; concatenating the second sequence of audio embeddings and the sequence of synthetic audio embeddings to produce a combined sequence of audio embeddings; and classifying the labeled training video as real or fake based on the combined sequence of audio embeddings and the combined sequence of image embeddings.
9. The method of claim 8, wherein training the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network further comprises computing a cross-entropy loss objective using label indicating whether the labeled training video is real or fake.
10. The method of claim 1, wherein the classifier comprises an audio mode patch reduction network, a visual mode patch reduction network, and a classifier head.
11. The method of claim 1, wherein a number of image tiles in the sequence of image tiles and a number of data segments in the sequence of data segments are determined based on a sampling frequency of the image data, a sampling frequency of the audio data, and a time duration of the input video.
12. The method of claim 1, further comprising: providing the trained model with a second input video; and classifying the second input video as real or fake using the trained model.
13. The method of claim 12, wherein the second input video comprises real audio data and AI-generated image data.
14. The method of claim 12, wherein the second input video comprises real image data and AI-generated audio data.
15. The method of claim 12, wherein the second input video comprises AI-generated image data and AI-generated audio data.
16. The method of claim 1, wherein the input video shows a human face.
17. A system for training a model comprising a visual encoder, an audio encoder, an audio-to-visual

(A2V) network, a visual-to-audio (V2A) network, and a classifier for classifying videos as real or fake, the system comprising one or more processors configured to: generate a sequence of image tiles from image data from an input video; generate a plurality of data segments representing audio data from the input video; generate a sequence of image embeddings based on the sequence of image tiles using the visual encoder; generate a sequence of audio embeddings based on the sequence of data segments using the audio encoder; transform, using the V2A network, a first subset of the sequence of image embeddings into one or more synthetic audio embeddings, wherein the first subset of the sequence of image embeddings corresponds to a first set of time points in the input video; transform, using the A2V network, a first subset of the sequence of audio embeddings into one or more synthetic image embeddings, wherein the first subset of the sequence of audio embeddings corresponds to a second set of time points in the input video complementary to the first set of time points; update the sequence of image embeddings by replacing a second subset of the sequence of image embeddings with the one or more synthetic image embeddings, wherein the second subset of the sequence of image embeddings corresponds to the second set of time points; update the sequence of audio embeddings by replacing a second subset of the sequence of audio embeddings with the one or more synthetic audio embeddings, wherein the second subset of the sequence of audio embeddings corresponds to the first set of time points; train the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated sequence of audio embeddings; train the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network, wherein the classifier is configured to receive image embeddings for the videos from the trained visual encoder, audio embeddings for the videos from the trained audio encoder, synthetic image embeddings for the videos from the trained A2V network, and synthetic audio embeddings for the videos from the trained V2A network.

18. A non-transitory computer readable storage medium storing instructions for training a model comprising a visual encoder, an audio encoder, an audio-to-visual (A2V) network, a visual-to-audio (V2A) network, and a classifier for classifying videos as real or fake that, when executed by one or more processors of a computer system, cause the computer system to: generate a sequence of image tiles from image data from an input video; generate a plurality of data segments representing audio data from the input video; generate a sequence of image embeddings based on the sequence of image tiles using the visual encoder; generate a sequence of audio embeddings based on the sequence of data segments using the audio encoder; transform, using the V2A network, a first subset of the sequence of image embeddings into one or more synthetic audio embeddings, wherein the first subset of the sequence of image embeddings corresponds to a first set of time points in the input video; transform, using the A2V network, a first subset of the sequence of audio embeddings into one or more synthetic image embeddings, wherein the first subset of the sequence of audio embeddings corresponds to a second set of time points in the input video complementary to the first set of time points; update the sequence of image embeddings by replacing a second subset of the sequence of image embeddings with the one or more synthetic image embeddings, wherein the second subset of the sequence of image embeddings corresponds to the second set of time points; update the sequence of audio embeddings by replacing a second subset of the sequence of audio embeddings with the one or more synthetic audio embeddings, wherein the second subset of the sequence of audio embeddings corresponds to the first set of time points; train the visual encoder, the audio encoder, the V2A network, and the A2V network based on the updated sequence of image embeddings and the updated sequence of audio embeddings; train the classifier to classify videos as real or fake using the trained visual encoder, the trained audio encoder, the trained V2A network, and the trained A2V network, wherein the classifier is configured to receive image embeddings for the videos from the trained visual encoder, audio embeddings for the videos from the trained audio encoder, synthetic image embeddings for the videos from the trained A2V network, and synthetic audio embeddings for the videos from the trained V2A network.

